Flash the clean Jetson system

Flash the clean Jetson system

1. Hardware connections
    1.1. Flash mode
    1.2. Mainboard connections
2. Software connection
3. Using SDK Manager
    3.1. Downloading SDK Manager
    3.2. Installing SDK Manager
    3.3. Launching and logging in to the account
    3.4. Mainboard selection
    3.5. STEP1
    3.6. STEP2
    3.7. STEP3
    3.8. STEP4
4. Booting the system
    4.1. Normal mode
    4.2. Booting the system
    4.3. System setup
    Common problems
        Device not detected

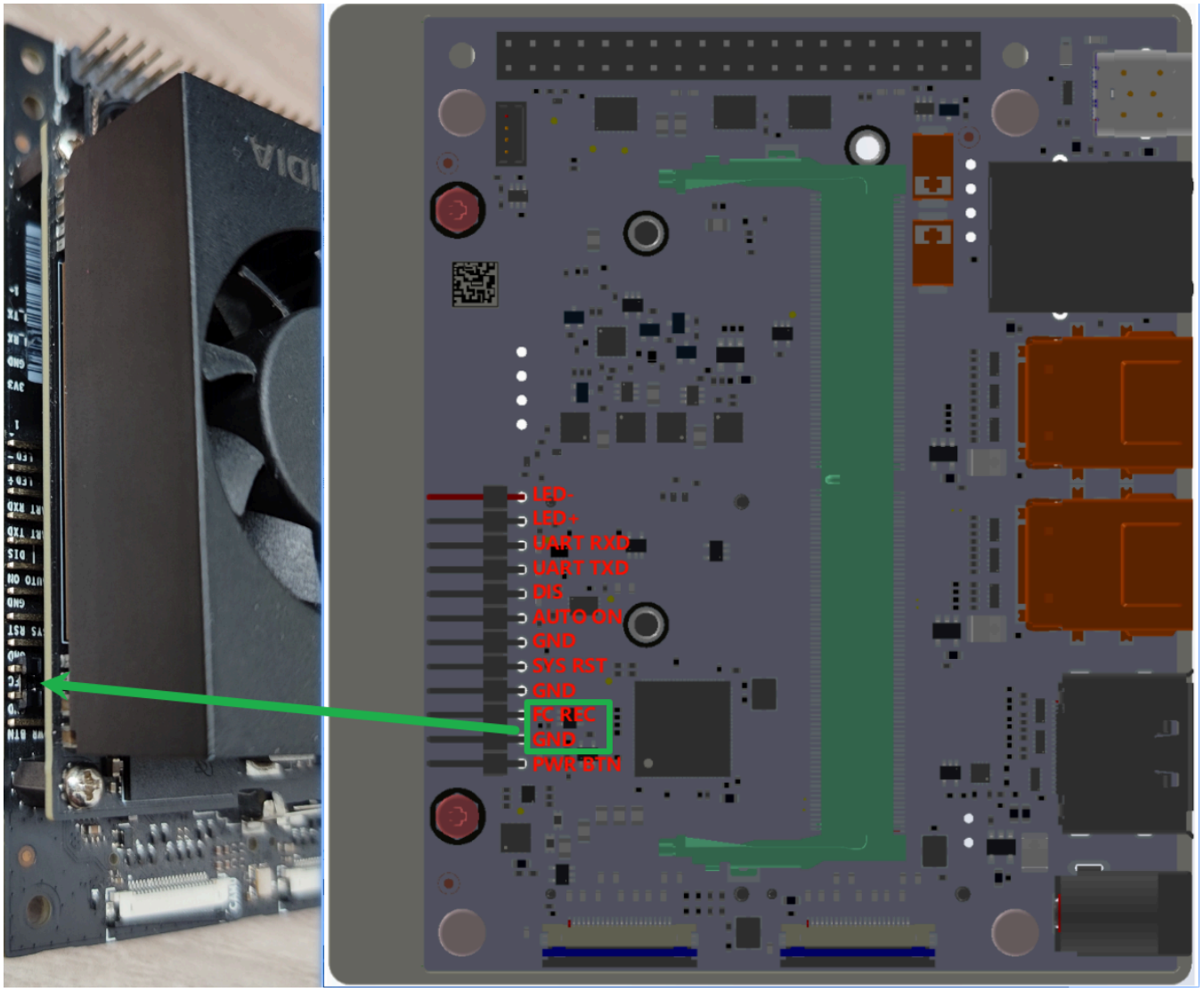
This tutorial demonstrates flashing the official NVIDIA system image on a Jetson Orin **official kit** (the SUB kit cannot activate SUPER mode in this way). It is not recommended for beginners!

Note: The SSD bundled with the product already contains the factory image, and the factory image includes the official NVIDIA system and the mainboard environment; do not proceed unless you specifically need the clean system!

1. Hardware connections

1.1. Flash mode

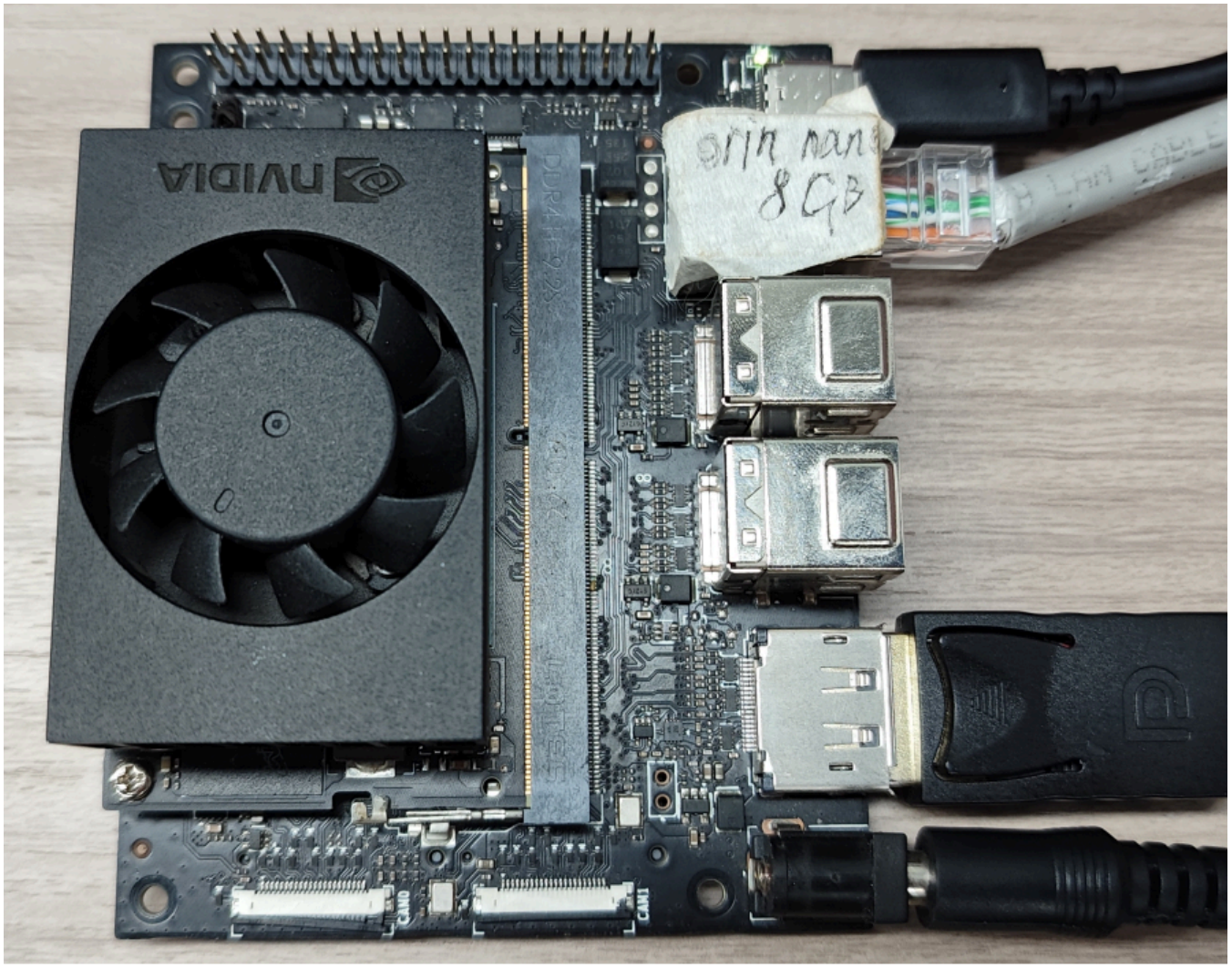
To enter flash mode, the Jetson Orin mainboard requires shorting `FC REC` and `GND` below the core board:



1.2. Mainboard connections

The illustration below uses Jetson Orin as an example; users of other versions can refer to it (the hardware interfaces and functional layout are identical)

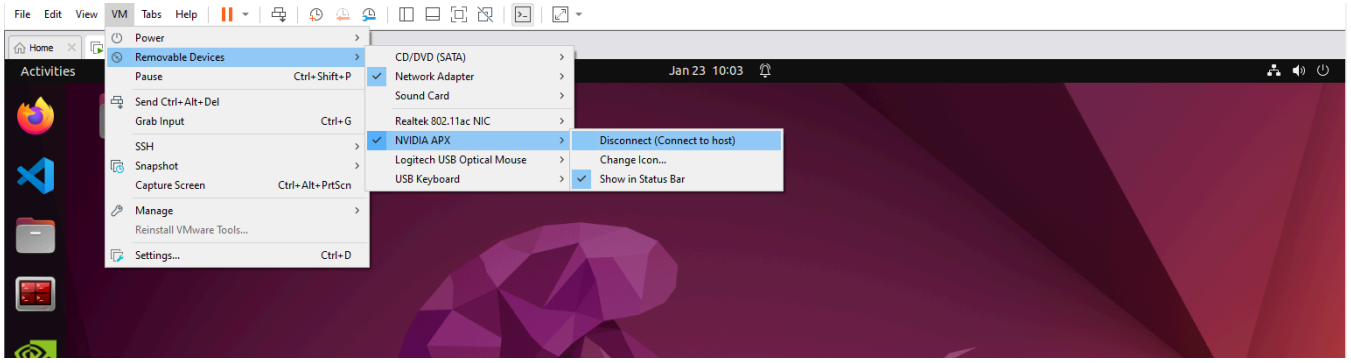
The Jetson Orin mainboard must be connected to a DC power adapter, a DP cable, an Ethernet cable and a Type C cable: connect the Type C cable to the computer



2. Software connection

This tutorial uses VMware Workstation 17 Pro to flash the system!

After starting the virtual machine, select in the menu bar: Virtual Machine → NVIDIA APX → confirm the connection status with the virtual machine



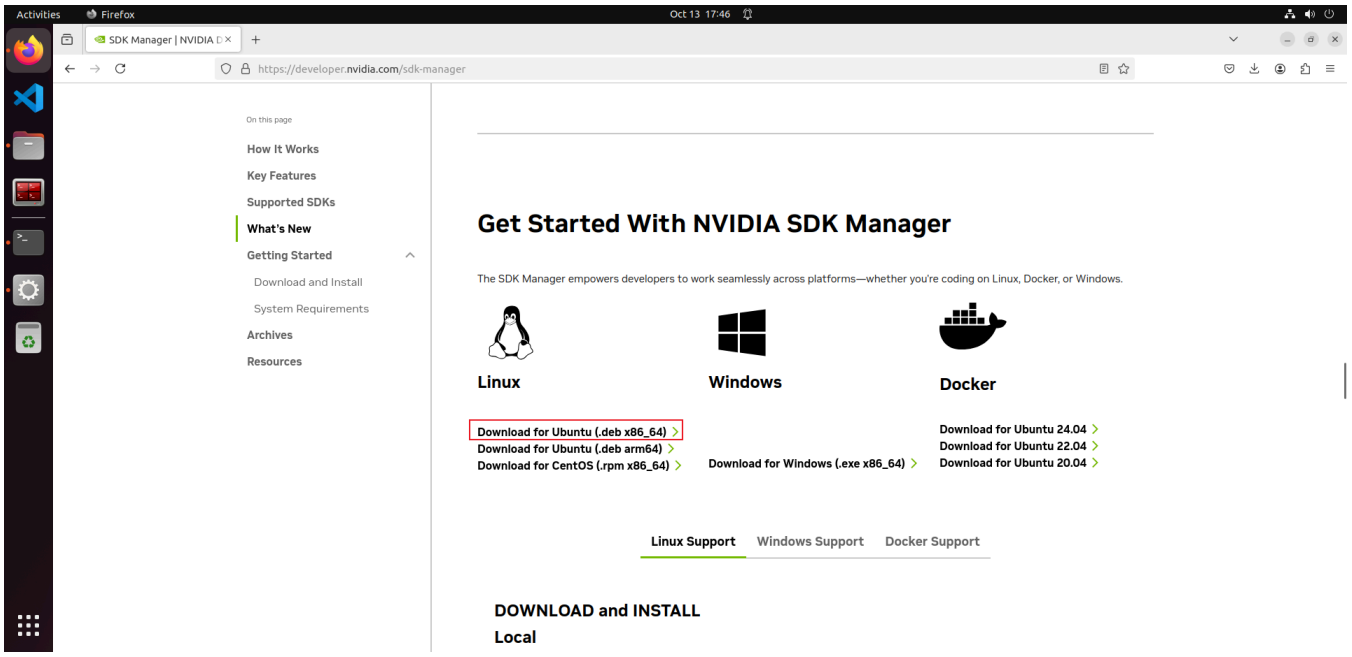
3. Using SDK Manager

Using SDK Manager requires registering an account and logging in beforehand; complete the registration and the login yourself!

```
You do not need to install SDK Manager on our virtual machine; if SDK Manager cannot be updated, you can reinstall it following the tutorial below
```

3.1. Downloading SDK Manager

Official website: https://developer.nvidia.com/sdk-manager



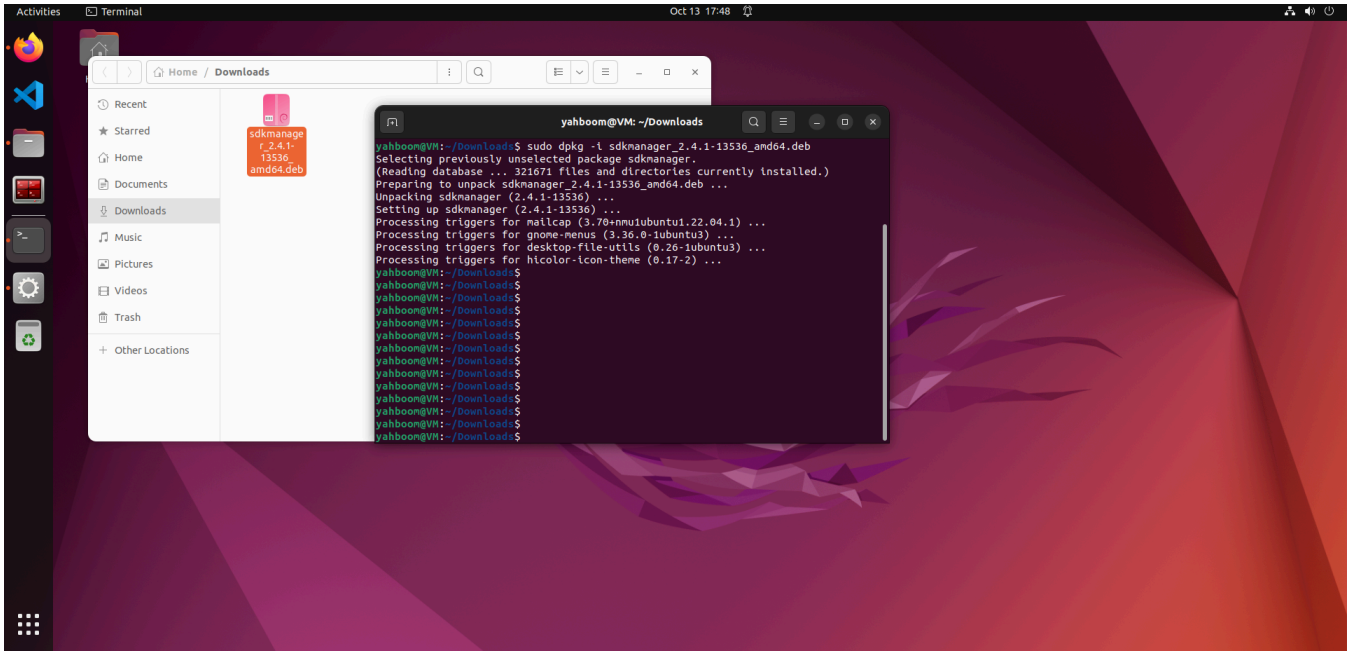
3.2. Installing SDK Manager

Enter the folder where the installation package is located and run the installation command; note that the file name in the command must match the actual installation package:

```
sudo dpkg -i sdkmanager_2.4.1-13536_amd64.deb
```

Fixing the dependencies: a clean system may report an error when installing sdkmanager; you can run the following command and then run the installation command again

```
sudo apt --fix-broken install -y
```

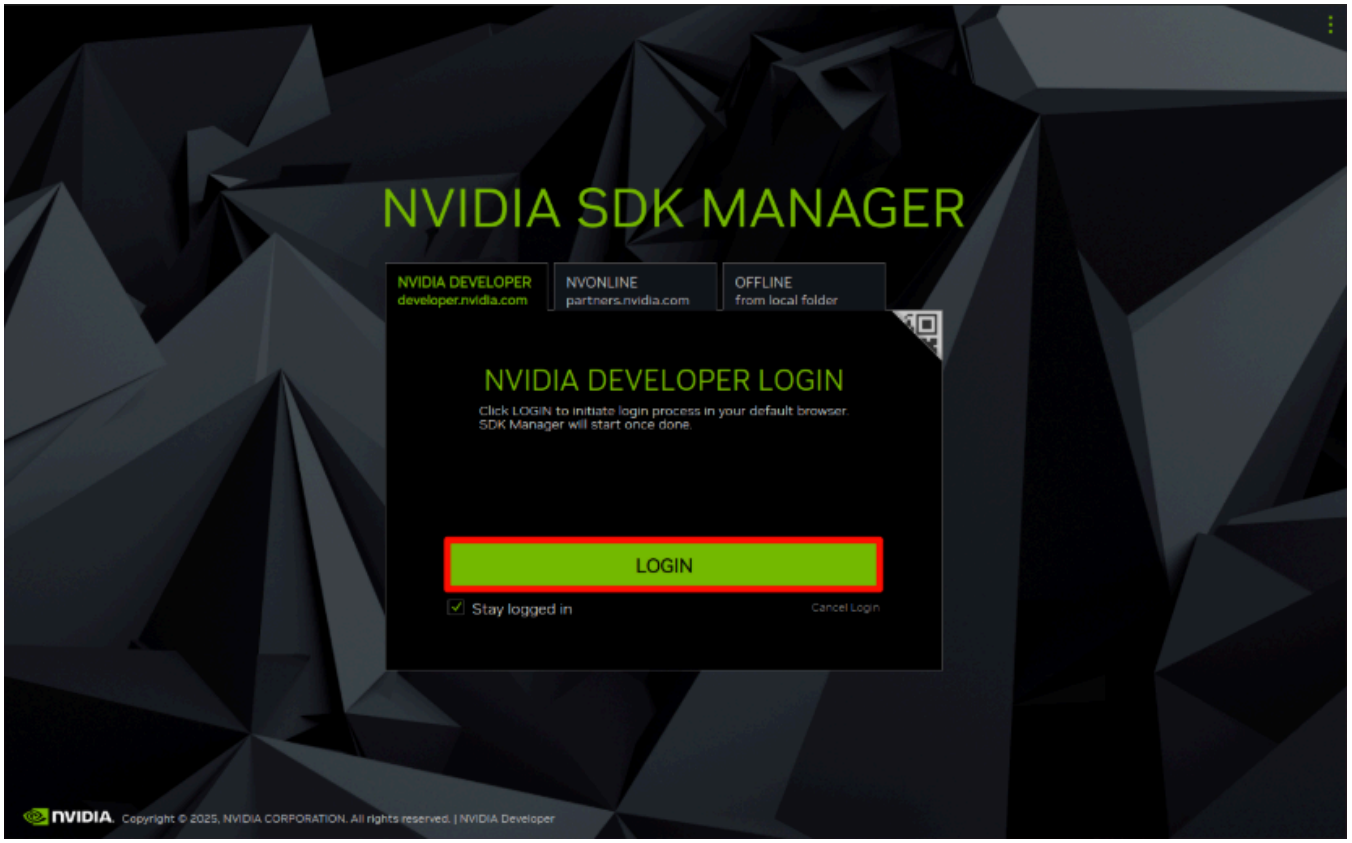


3.3. Launching and logging in to the account

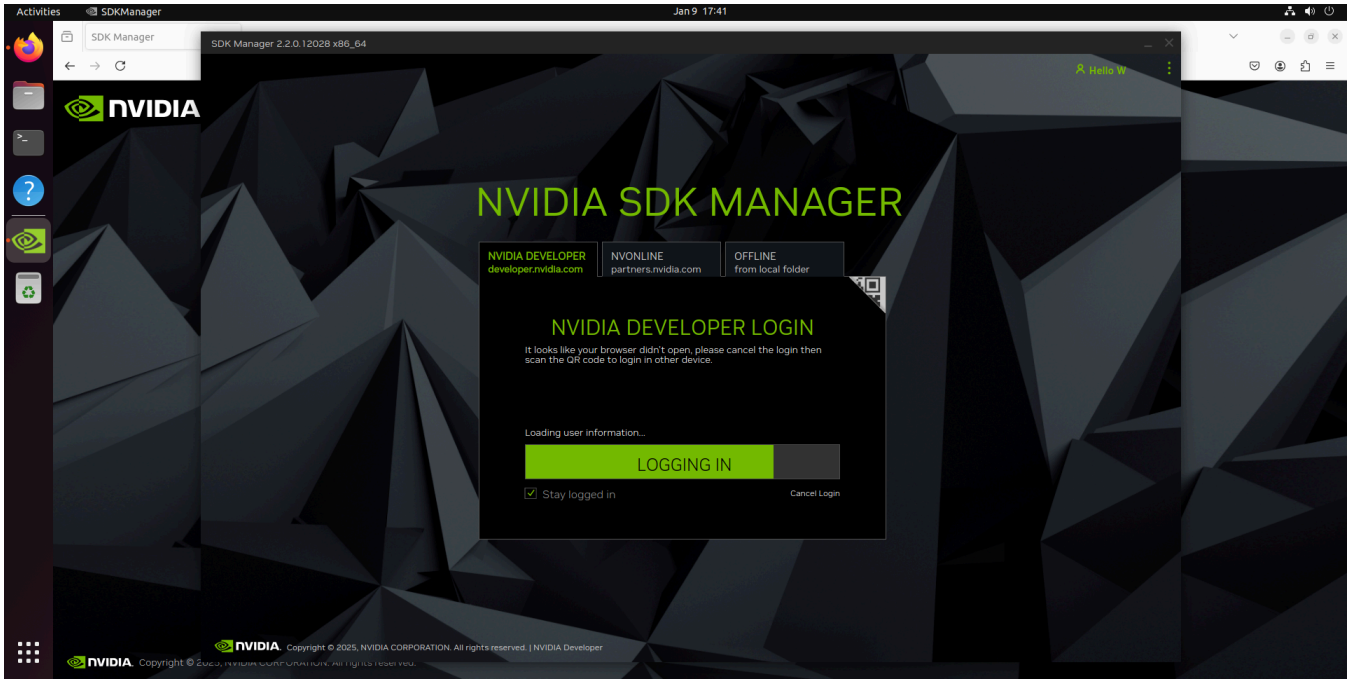
Find the `SDK Manager` application icon in the system application menu, or launch it by entering the following command in a terminal:

```
sdkmanager
# To flash an older JetPack version, use the command: sdkmanager --archived-versions
```

After clicking Login, the login page of the browser pops up automatically:

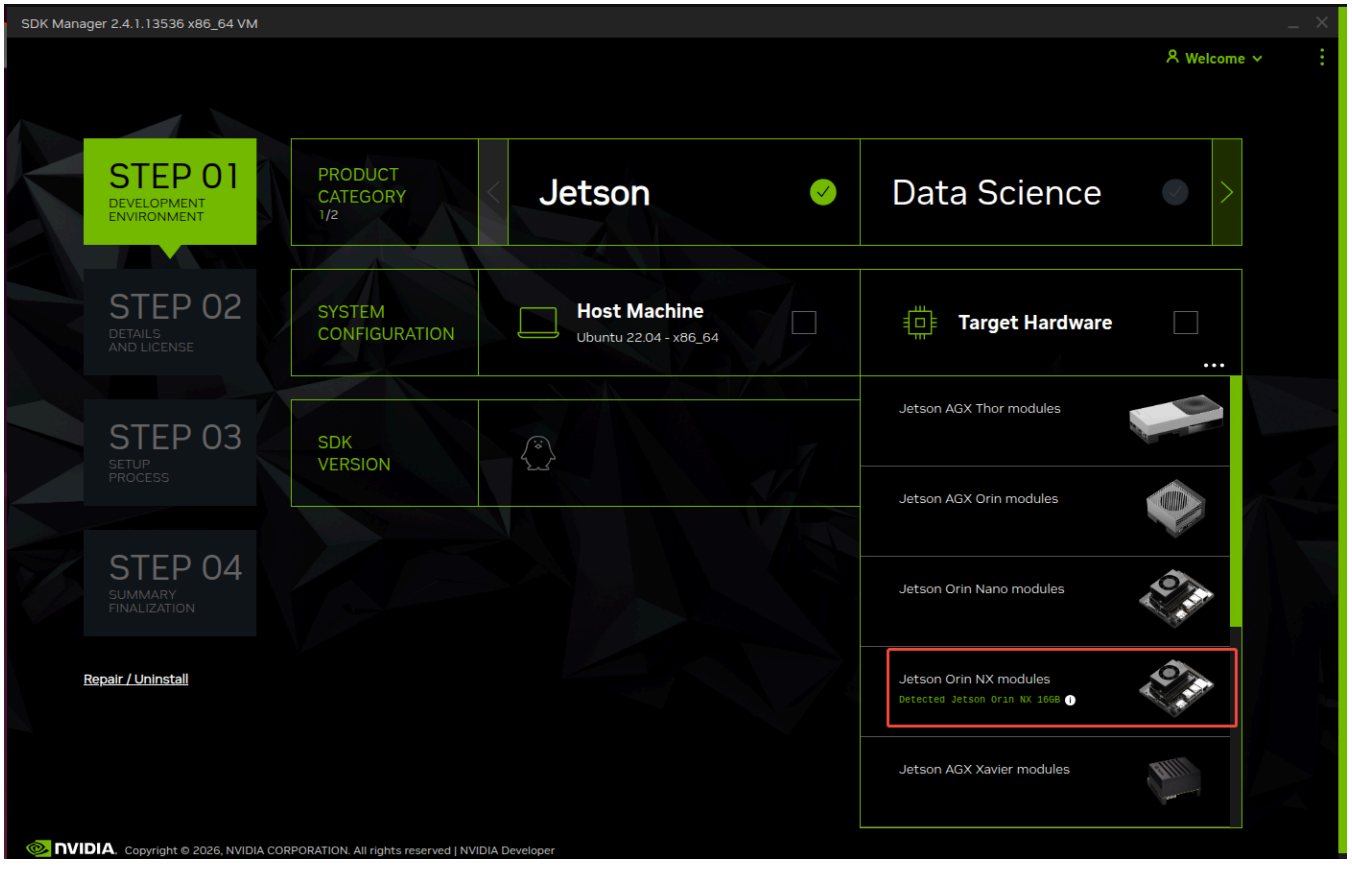


Enter the account and password you registered earlier to enter the software:



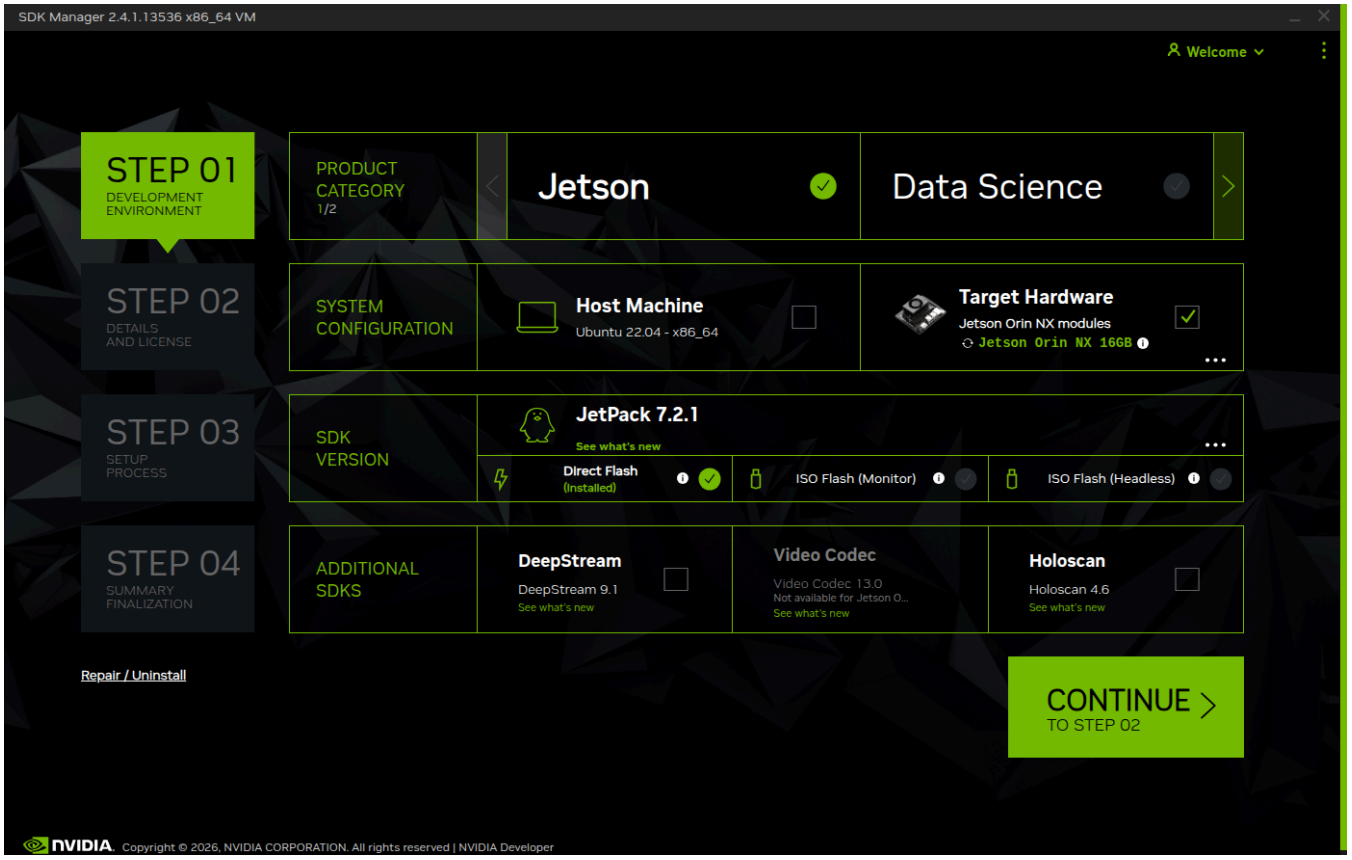
3.4. Mainboard selection

Select according to your mainboard model: for the official kit choose `Jetson Orin NX`



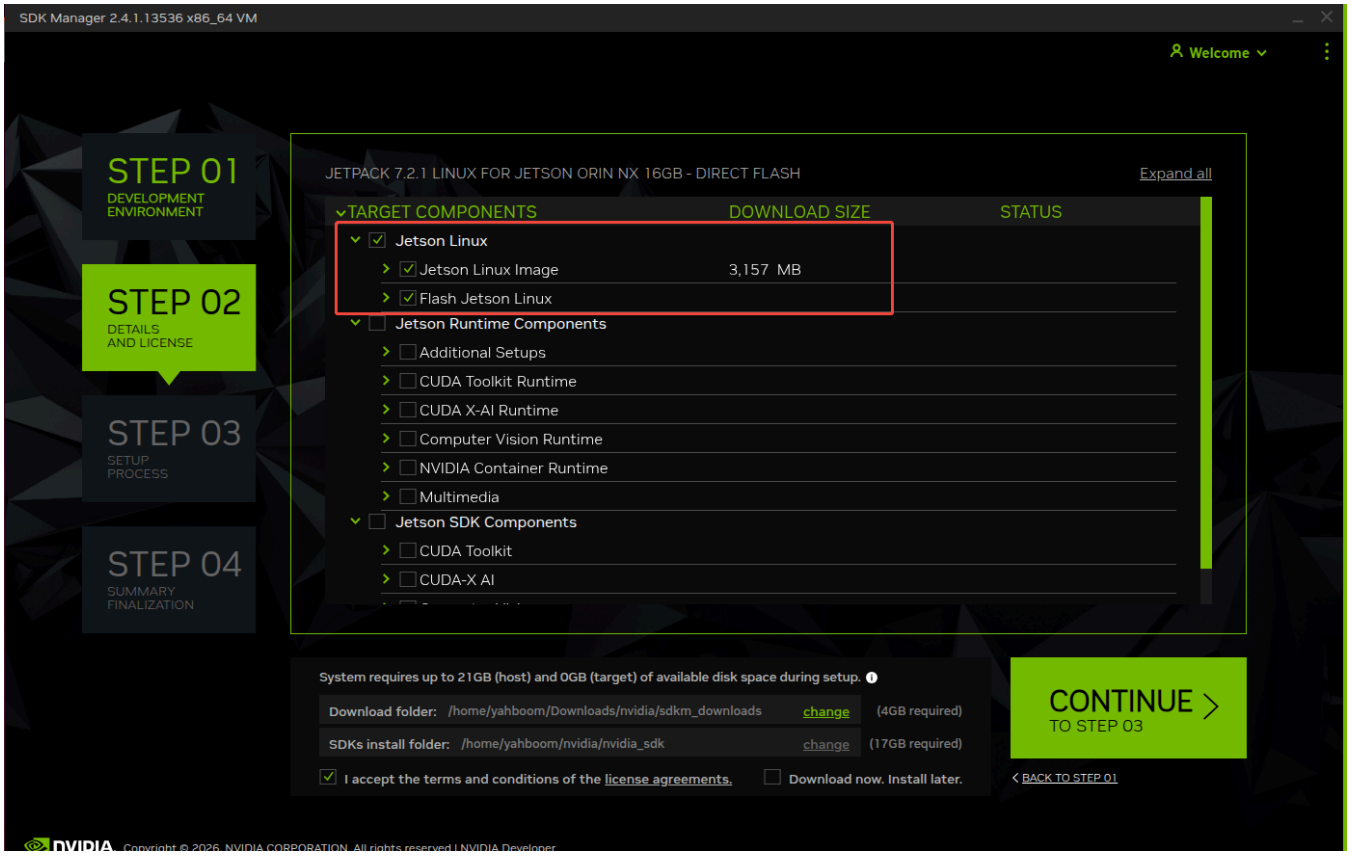
3.5. STEP1

Tick the options as shown below; after confirming that the selections match, click `CONTINUE`:

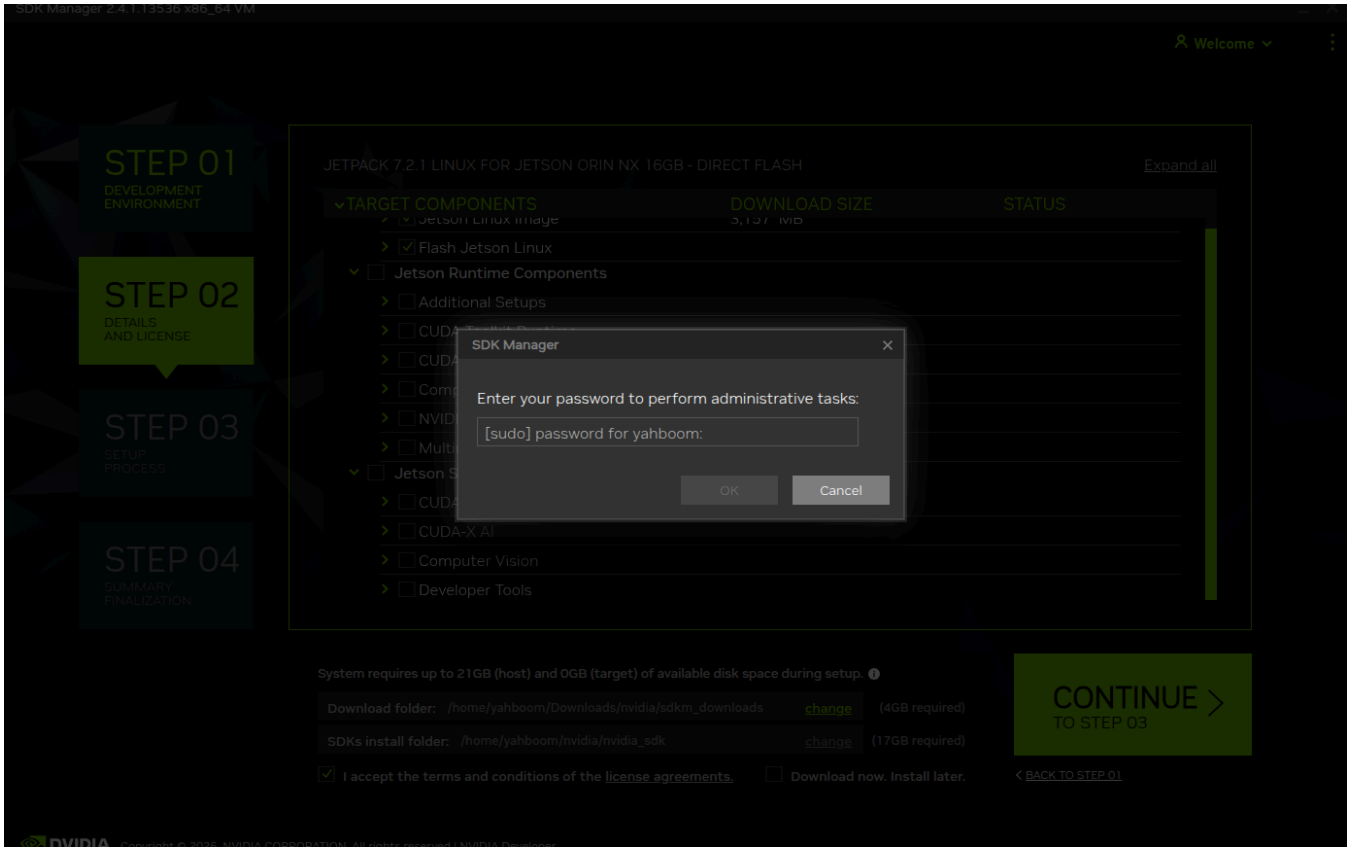


3.6. STEP2

Tick the options as shown below; after confirming that the selections match, click `CONTINUE`:



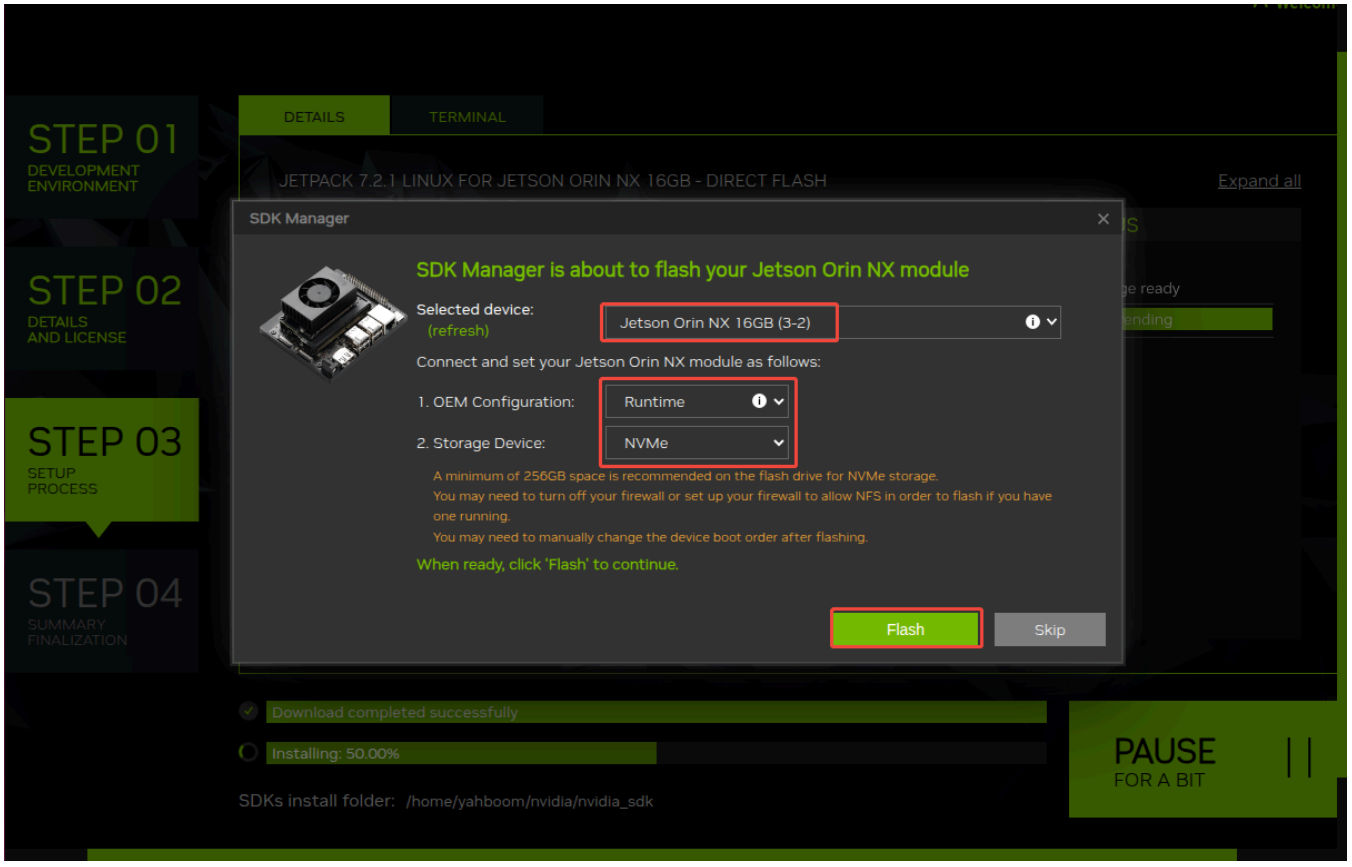
Enter the virtual machine password: yahboom; if it is your own virtual machine, enter your own password



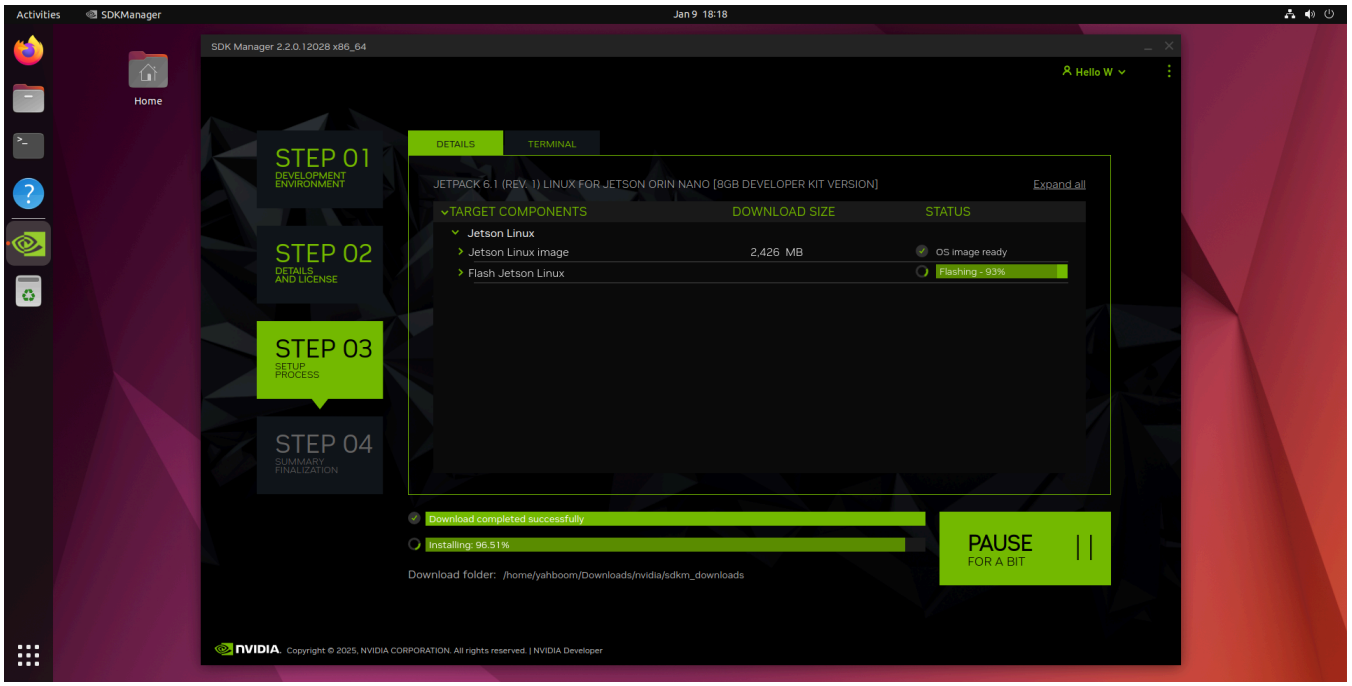
3.7. STEP3

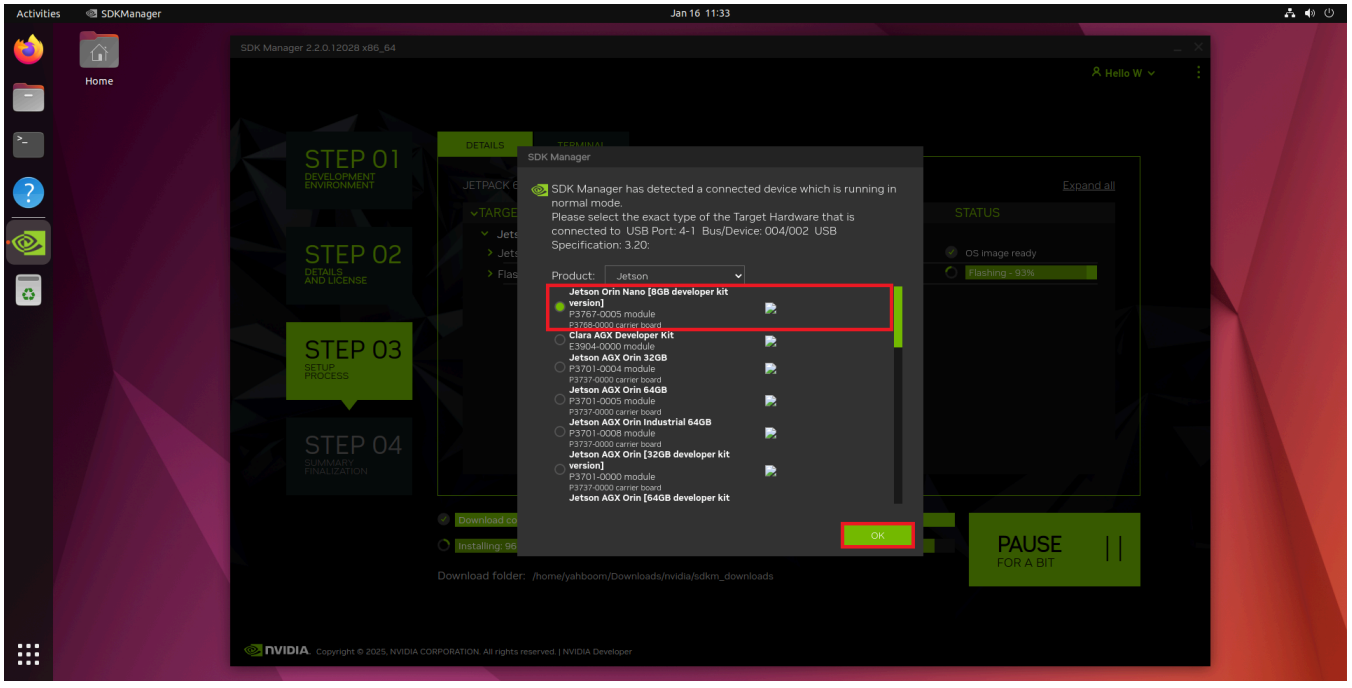
Wait for the system to download and install: when the installation reaches 50%, the user must set the options

Select the mainboard model, the OEM configuration and the system storage medium:

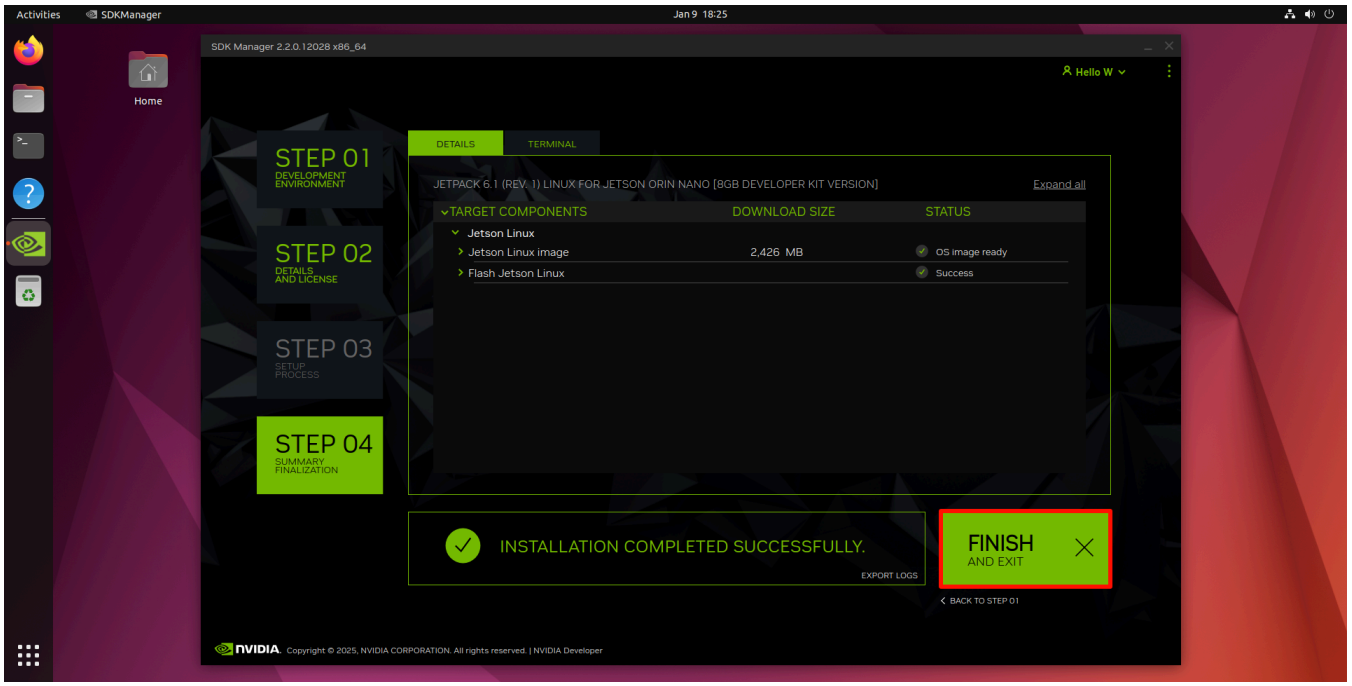


Throughout the process the virtual machine automatically controls the connection and disconnection of the device, but once the progress passes 70% you need to watch the progress of the system flashing in SDK Manager; if the connection is dropped and the virtual machine does not reconnect automatically, manually reconnect the mainboard to the virtual machine in time:





3.8. STEP4



4. Booting the system

After the system has been flashed successfully with SDK Manager, disconnect the DC power and the Type C cable of the mainboard, remove the shorting jumper between FC REC and GND below the core board while the power is off, then connect a monitor, a keyboard and a mouse to the mainboard to make the initial system setup easier, and finally power it with the DC power adapter!

4.1. Normal mode

Remove the `FC REC` and `GND` jumper caps that were shorted when flashing the Jetson Orin mainboard.

4.2. Booting the system

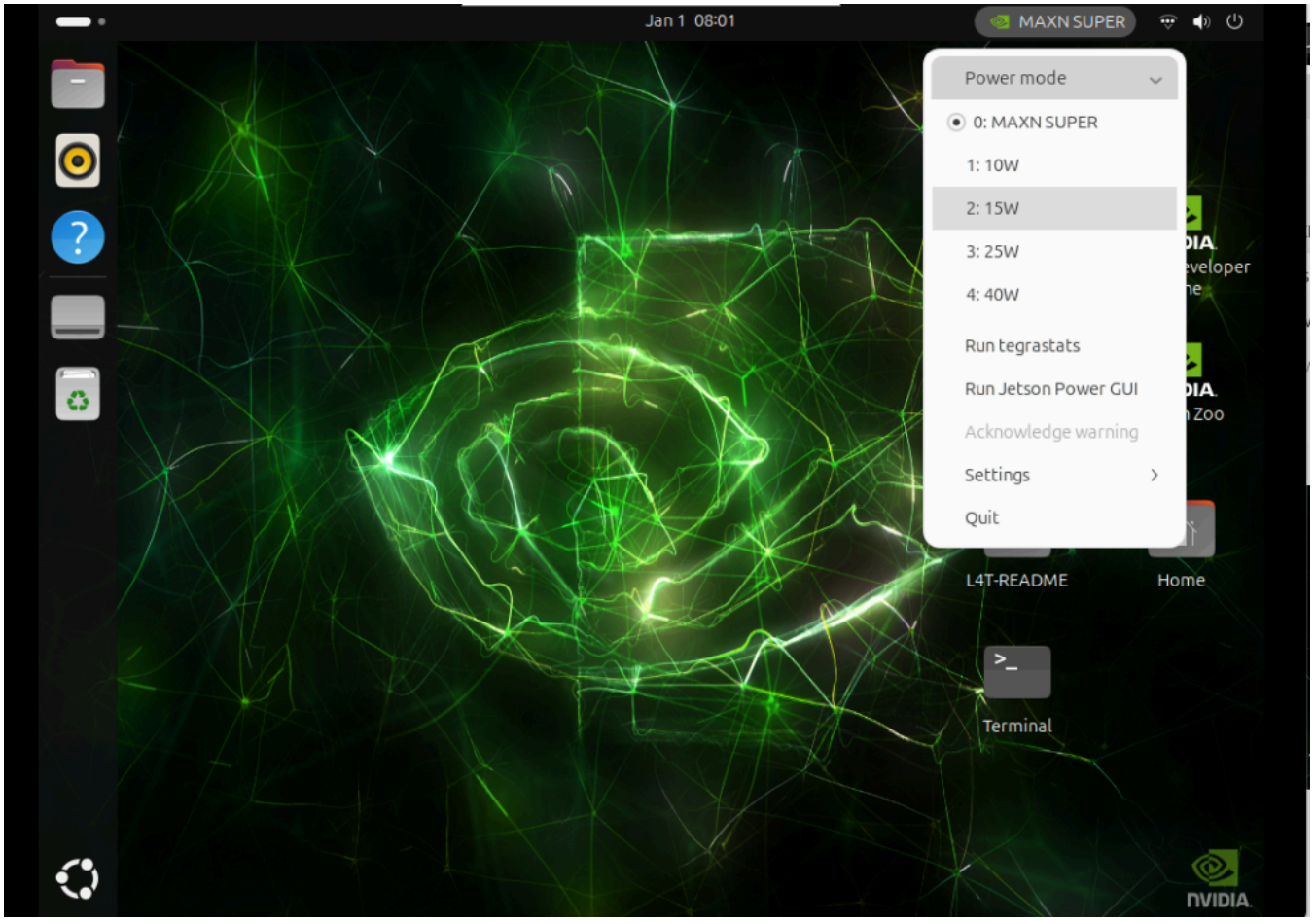
Connect a DP cable, an Ethernet cable and a Type C cable to the Jetson mainboard, and finally power the Jetson mainboard with the DC power adapter.

```
The Jetson Orin series mainboards require a display to show the picture properly and to perform VNC remote access
```

4.3. System setup

After the system boots, the user can set the system options according to the on-screen prompts; the process is fairly simple, so it is not demonstrated here. Finally you enter the system:


Jan 1 08:02
MAXN SUPER
NVIDIA Jetson Community Proj…
NVIDIA Jetson Developer Zone
NVIDIA Jetson Support Forums
NVIDIA Jetson Zoo
L4T-README
Home
Terminal
NVIDIA


Jan 1 08:01
MAXN SUPER
Power mode
0: MAXN SUPER
1: 10W
2: 15W
3: 25W
4: 40W
Run tegrastats
Run Jetson Power GUI
Acknowledge warning
Settings
Quit
L4T-README
Home
Terminal
NVIDIA

Common problems

Device not detected

You can disconnect the DC power and the Type C cable of the mainboard, and then power it on again and connect it to the virtual machine!